Practice assignment 10 - Not Graded

This assignment will not be graded and is only for practice.

1 point

Match the functions of two variables in Column A with their directions in which the directional derivatives at (0,0) is maximized given in Column B.

- [ ] (1) ⟶ (v)(1) ⟶ (v)
- [ ] (1) ⟶ (iii)(1) ⟶ (iii)
- [ ] (2) ⟶ (iv)(2) ⟶ (iv)
- [ ] (3) ⟶ (iv)(3) ⟶ (iv)
- [ ] (2) ⟶ (vi)(2) ⟶ (vi)
- [ ] (4) ⟶ (v)(4) ⟶ (v)
- [ ] (3) ⟶ (vi)(3) ⟶ (vi)
- [ ] (4) ⟶ (i)(4) ⟶ (i)

No, the answer is incorrect.

Score: 0

Accepted Answers:

(1) ⟶ (iii)(1) ⟶ (iii)

(2) ⟶ (iv)(2) ⟶ (iv)

(4) ⟶ (v)(4) ⟶ (v)

1 point

Which of the following statements is(are) correct?

- [ ] Maximum directional derivative at (0,0) for the function $f(x, y) = |x| + |y|$f(x, y) = |x| + |y| is 1.
- [ ] The graph of the linear approximation to a function $f(x, y)$f(x, y) at a given point PP is the tangent plane to $f(x, y)$f(x, y) at the point PP.
- [ ] A tangent plane to a function $f(x, y)$f(x, y) exists at every point.
- [ ] All directional derivatives of a function $f(x, y)$f(x, y) at a point PP are 0 if the tangent plane is parallel to the XYXY -plane.

No, the answer is incorrect.

Score: 0

Accepted Answers:

The graph of the linear approximation to a function $f(x, y)$f(x, y) at a given point PP is the tangent

plane to $f(x, y)$f(x, y) at the point PP.
All directional derivatives of a function $f(x, y)$f(x, y) at a point PP are 0 if the tangent plane is parallel to the XYXY -plane.
1 point
Which of the following is(are) critical points of the scalar valued multivariable function $f(x, y) = 6x^2 - 2x^3 + 3y^2 - 6xy$f(x, y) = 6x2 − 2x3 + 3y2 − 6xy?
☐ (1,1)
☐ (0,0)
☐ (1,-1)
☐ No critical points.
No, the answer is incorrect.
Score: 0
Accepted Answers:
(1,1)
(0,0)

Consider two functions f_1, f_2 f1, f2 defined as

$$f_1(x, y, z) = \ln(x^2 + y^2 + 2z^2)\text{f1(x, y, z)} = \ln(\text{x2} + \text{y2} + 2\text{z2})$$

and

$$f_2(x, y, z) = \tan^{-1}(x^2 + y^2 + 2z^2).\text{f2(x, y, z)} = \tan{-1}(\text{x2} + \text{y2} + 2\text{z2}).$$

If $L_{f_1}(x, y, z)$Lf1(x, y, z) and $L_{f_2}(x, y, z)$Lf2(x, y, z) are the linear approximation of $f_1(x, y, z)$f1(x, y, z) and $f_2(x, y, z)$f2(x, y, z) at point (1, -1, 1), then find the value of $85(\nabla L_{f_1}(x, y, z) \cdot \nabla L_{f_2}(x, y, z))$85(∇Lf1(x, y, z) · ∇Lf2(x, y, z)).
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 30.0
1 point

$z = f(x, y) = \frac{-7}{6}x^2 + \frac{1}{3}y^2 + \frac{11}{6}$z = f(x, y) = 6−7x2 + 31y2 + 611 represents a surface. Let (x_0, y_0, z_0)(x0, y0, z0) be the point where the tangent plane to the given surface is parallel to the plane $7x - 8y + 3z + 2 = 0$7x − 8y + 3z + 2 = 0. Find the value of $11x_0 + 5y_0 + 4z_0$11x0 + 5y0 + 4z0.
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 55
1 point

Find the number of points where the tangent plane to the function $f(x, y) = x^3 + y^3 + 3x^2 - 9x - 12y + 20$f(x, y) = x3 + y3 + 3x2 − 9x − 12y + 20 at the points is parallel to the XYXY -plane.
No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 4
1 point

- Suppose $ax + by + cz = d_1$ ax + by + cz = d1 is a tangent plane to the function $f(x, y) = x^3 + 3y^3 + 9xy$f(x, y) = x3 + 3y3 + 9xy at (-1,2) and $ax + by + cz = d_2$ ax + by + cz = d2 is a tangent plane to the same function $f(x, y)$f(x, y) at $(\alpha, 1)$(α, 1).

  Use the above information to answer questions 7-8:

1 point

Which of the following options is(are) true?

- [ ] $\alpha = -2$α = −2
- [ ] $\alpha = 2$α = 2
- [ ] There are only two possible candidates for αα.
- [ ] $|d_2 - d_1| = 12$|d2 − d1| = 12

No, the answer is incorrect.
Score: 0
Accepted Answers:
$\alpha = 2$α = 2
$|d_2 - d_1| = 12$|d2 − d1| = 12

Find the value of $\frac{2a+b+c}{d_1+d_2}$d1+d22a+b+c.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1

1 point

- Let $L(x, y)$L(x, y) be the linear approximation to $f(x, y) = xe^y + ln(x + y)$f(x, y) = xey + ln(x + y) at point (1,0).

  Use the above information to answer questions 9-10:

Suppose $L(x, y) = ax + by + c$L(x, y) = ax + by + c, where $a, b, c \in R$a, b, c ∈ R. Find the value of $a + b - c$a + b − c.

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 5

1 point

Find the approximate value of the function $f(x, y)$f(x, y) at (1.1,0.1) using the linear approximation function $L(x, y)$L(x, y).

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 1.4

1 point

Check Answers

Your score is: 0/10